The Evolution of Death

Understanding the Hydra Effect in Spatial Predator-Prey Models

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Literature

There exists a critical death rate that maximizes population count, balancing altruistic suicide and evolutionary suicide

Altruistic suicide - "suicide fostering the survival of others"

Evolutionary suicide - "an evolutionary process in which a viable population adapts in such a way that it can no longer persist"

Hypotheses

1. Universality: this critical point is intrinsic to the predator-prey relation, not the model or parameter values

2. Self-organization: prey evolve toward this critical point through natural selection

REFERENCES

Evolution of suicide as a defence strategy against pathogens in a spatially structured environment - Berngruber - 2013 - Ecology Letters - Wiley Online Library
Small-scale spatial structure affects predator-prey dynamics and coexistence | bioRxiv
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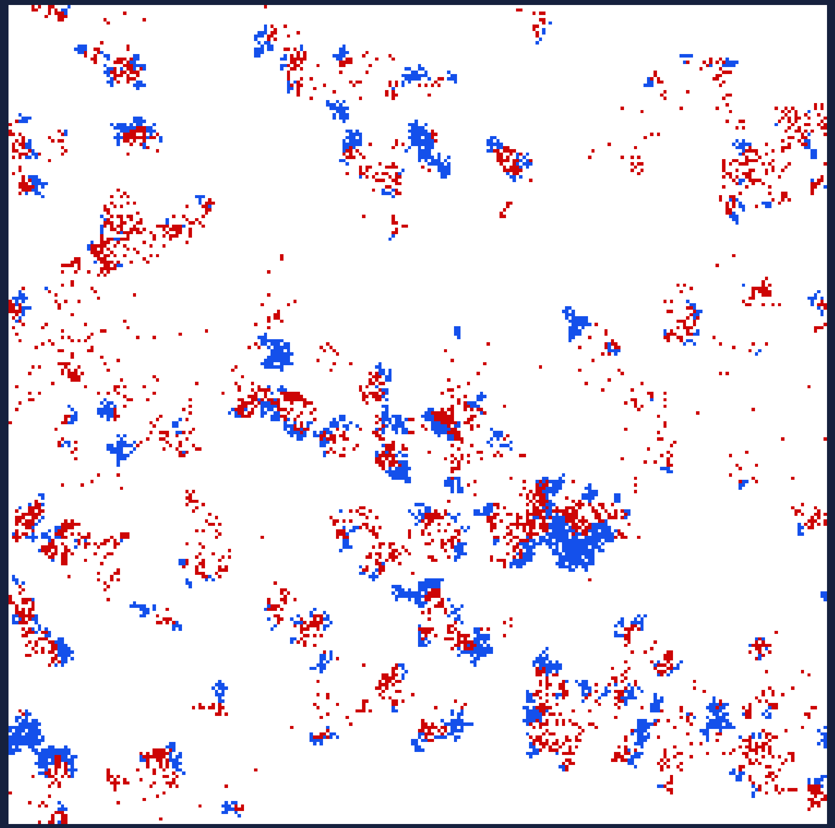
https://link.springer.com/chapter/10.1007/3-540-44811-X_12

Neighbourhood: Moore
Prey birth: 0.2
Predator birth: 0.8
Predator death: 0.05

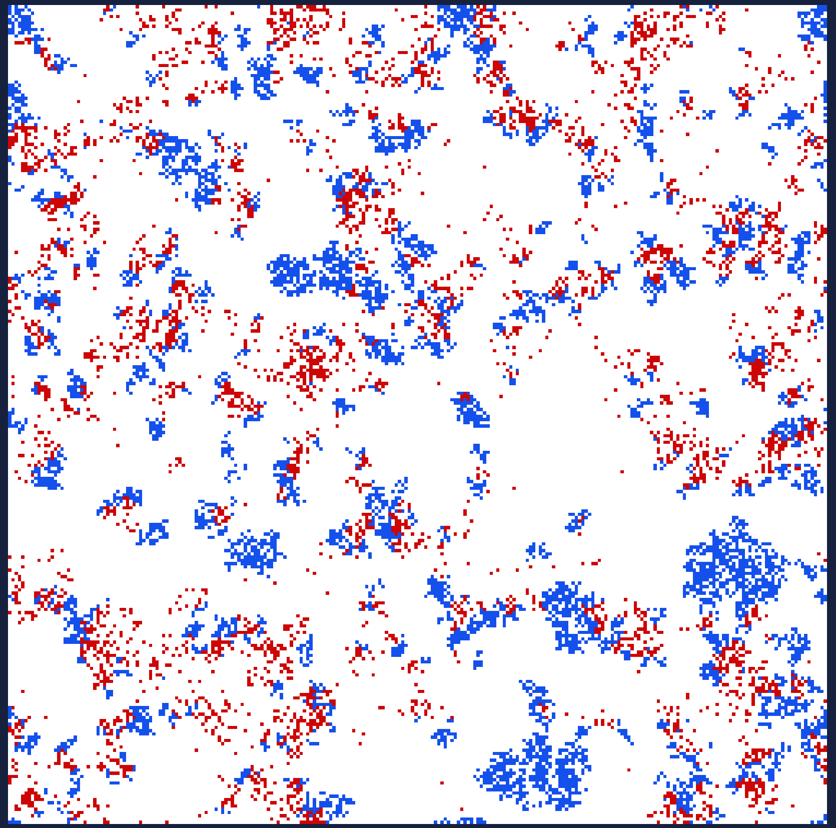
Experimental Setup

Time step: 1000
Grid: 250x250
Prey: Blue
Predator: Red

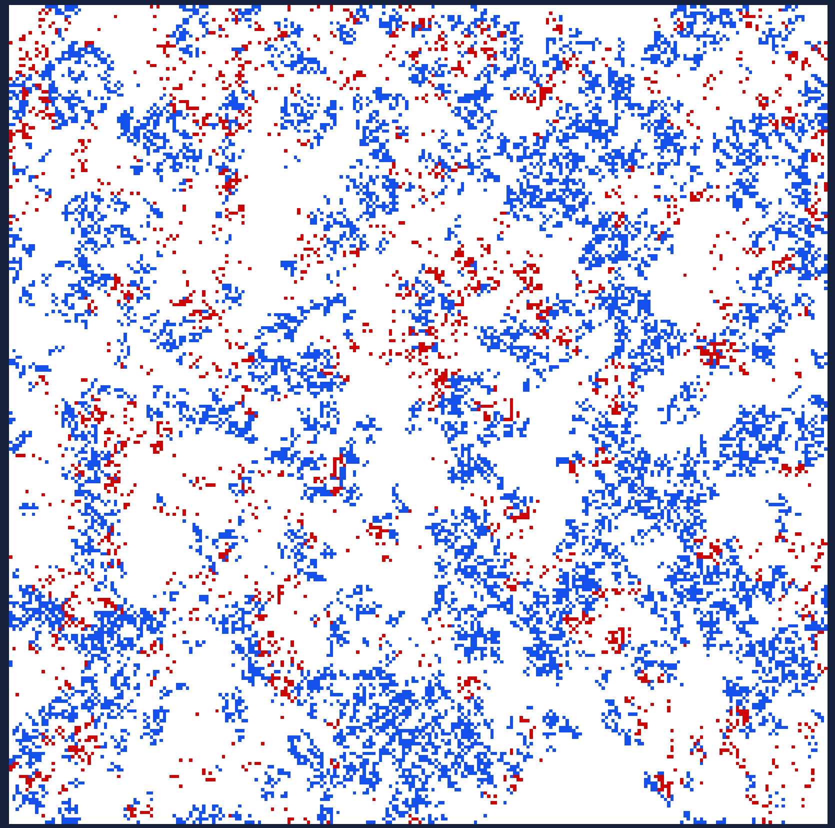
Prey death rate: 0.02



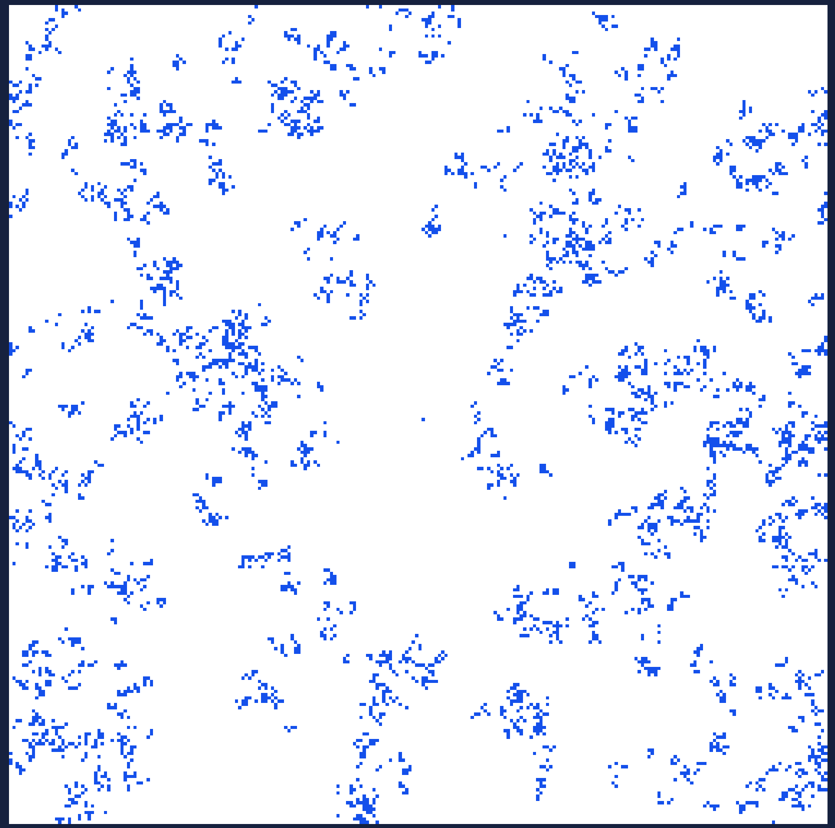
Prey death rate: 0.06



Prey death rate: 0.10



Prey death rate: 0.14



Phase Transition

- Hydra effect: increasing prey mortality lead to increase in prey
- Sweep over prey death rate
  - Low mortality: coexistence
  - Hydra region: prey increase, disrupted predator population
  - Extinction threshold: prey mortality exceeds reproduction capacity

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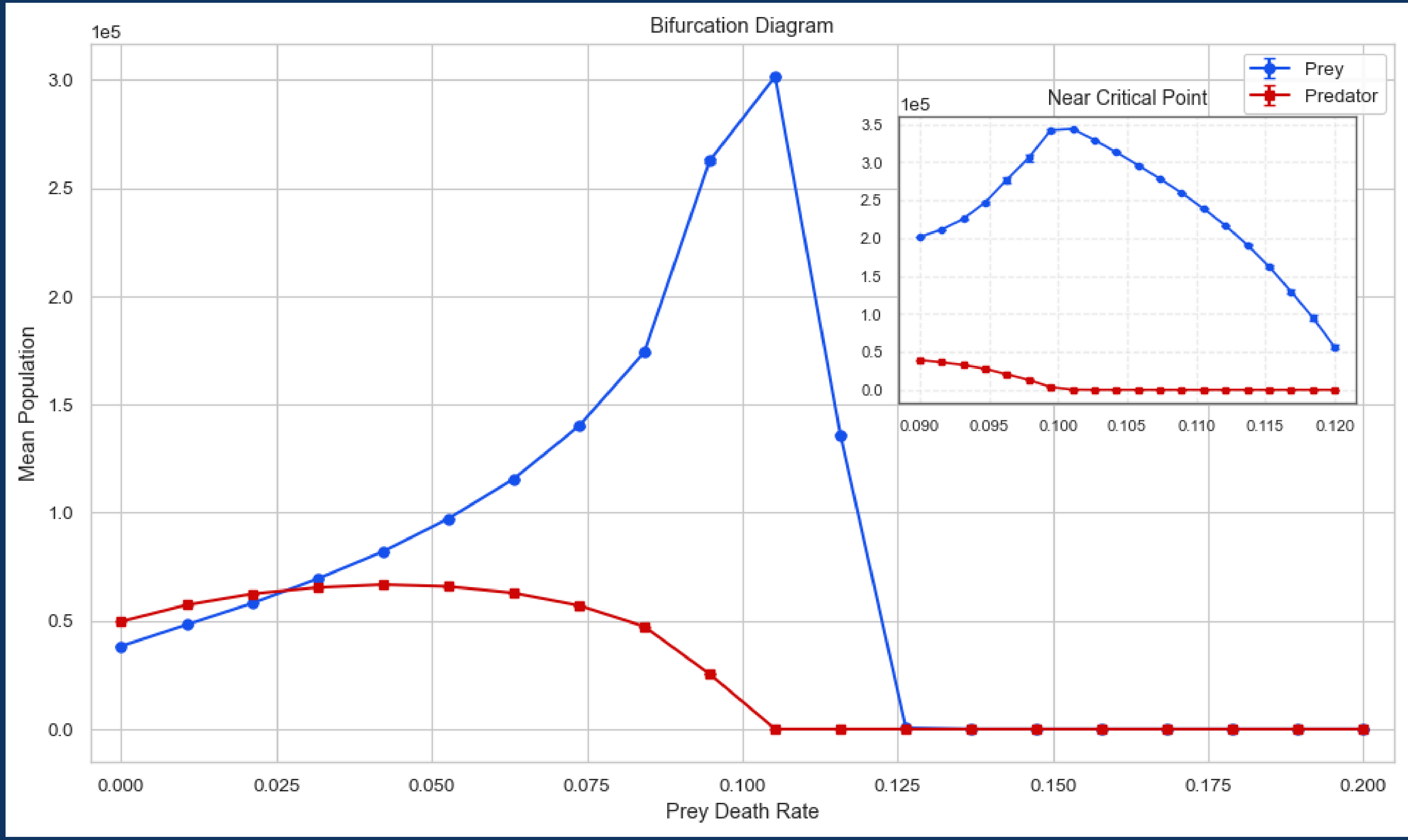


Criticality Test

Test statistic: Log-likelihood ratio (R)

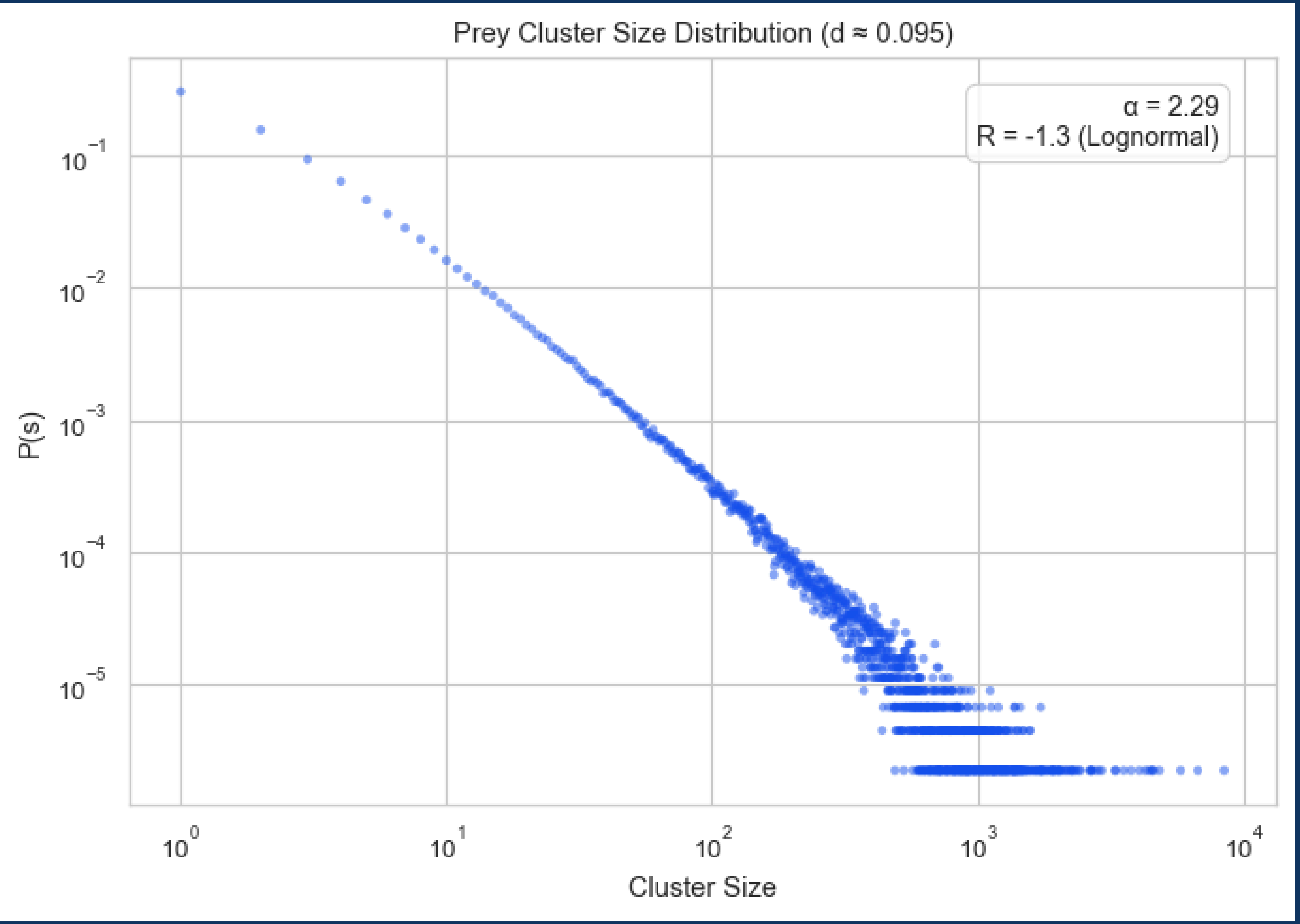
$$R = \mathcal{L}_{power-law} - \mathcal{L}_{lognormal}$$

$R > 0$: Power-law is a better fit

$R < 0$: Lognormal is a better fit

$|R| < 1$: Inconclusive (fits are comparable)

$\alpha = 2.29$: fitted power-law exponent



Finite-size scaling

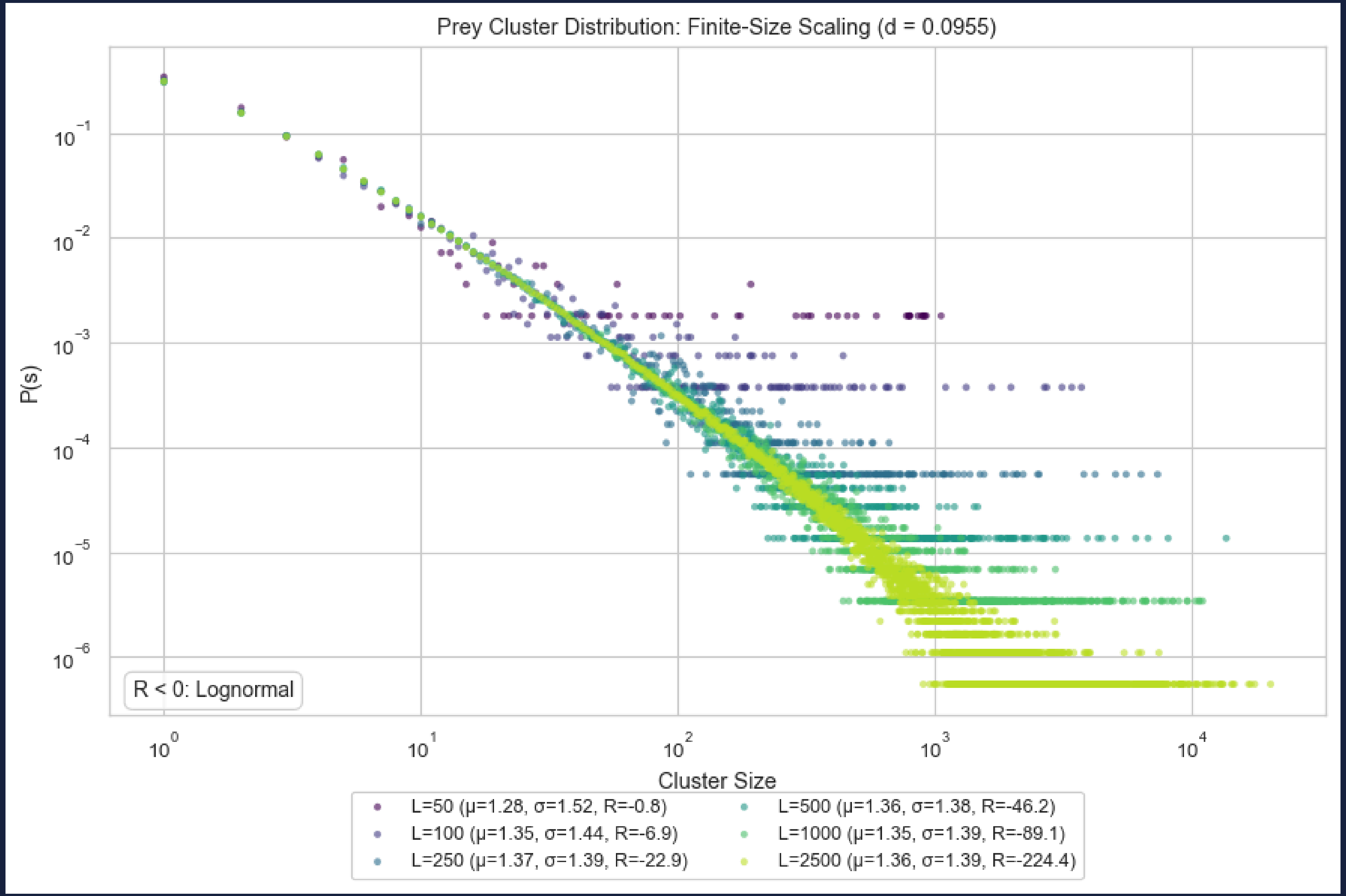
Lognormal distribution across grid sizes (L):

$$P(s) = \frac{1}{s\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln s - \mu)^2}{2\sigma^2}\right)$$

$\mu = \text{mean of } \ln(s)$ (log-scale location)

$\sigma = \text{standard deviation of } \ln(s)$ (log-scale spread)

As L increases, R becomes more negative



Sensitivity: sweep setup

Parameter Space (4D) :

- Prey birth(x11)/death(21)
- Predator birth(11)/death(11)

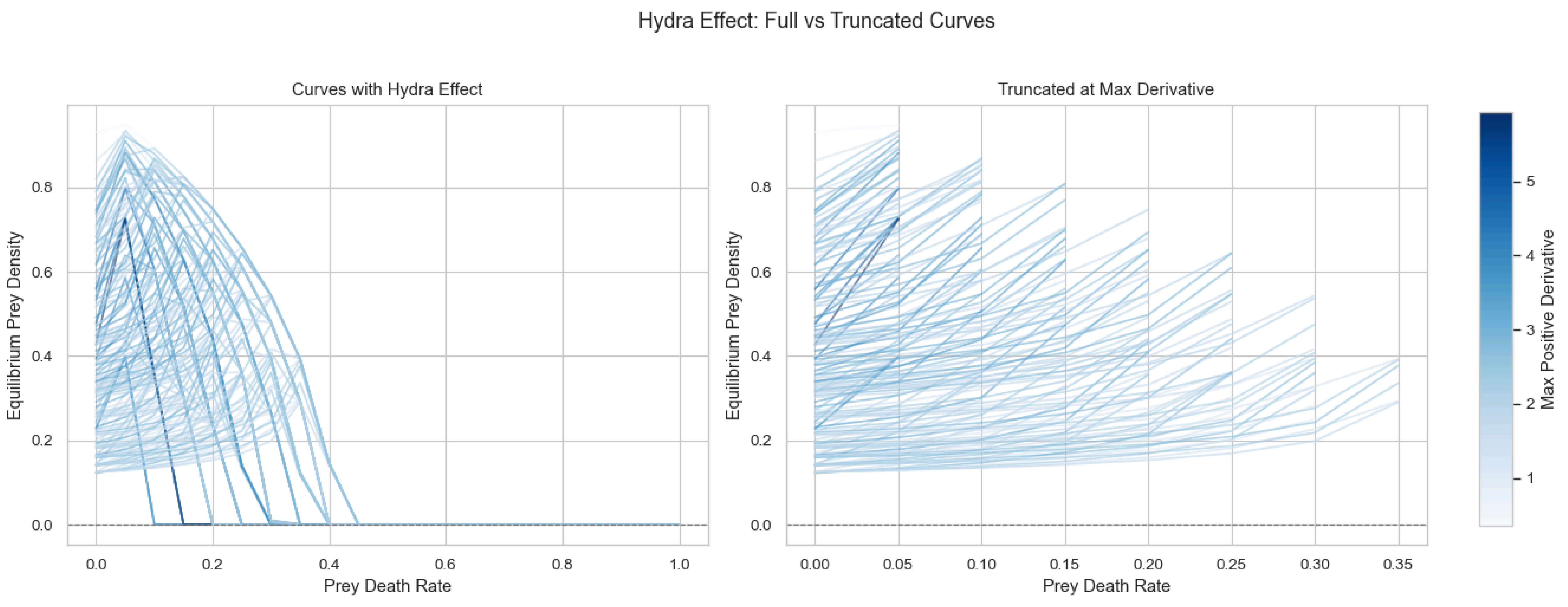
Hydra Region Detection:

$$\frac{d(\text{prey density})}{d(\text{prey death})} > 0$$

Quadratic Fitting:

$$\rho(d_1) = a \cdot d_1^2 + b \cdot d_1 + c$$

$$d_1 = \text{prey death rate}$$

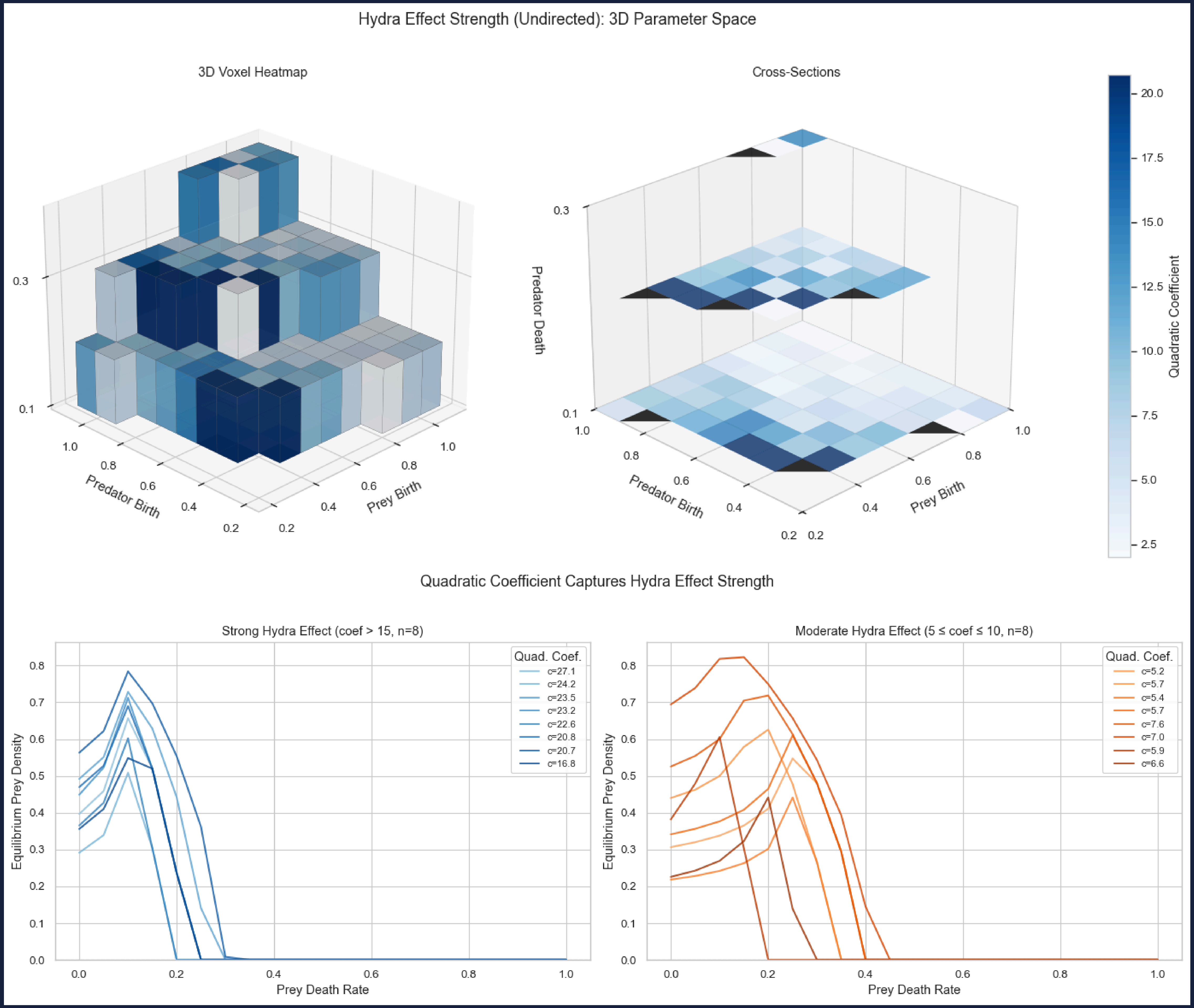


Undirected sweep

Undirected Hunting:

- Prey tries to reproduce in all cells
- Predator doesn't check for prey

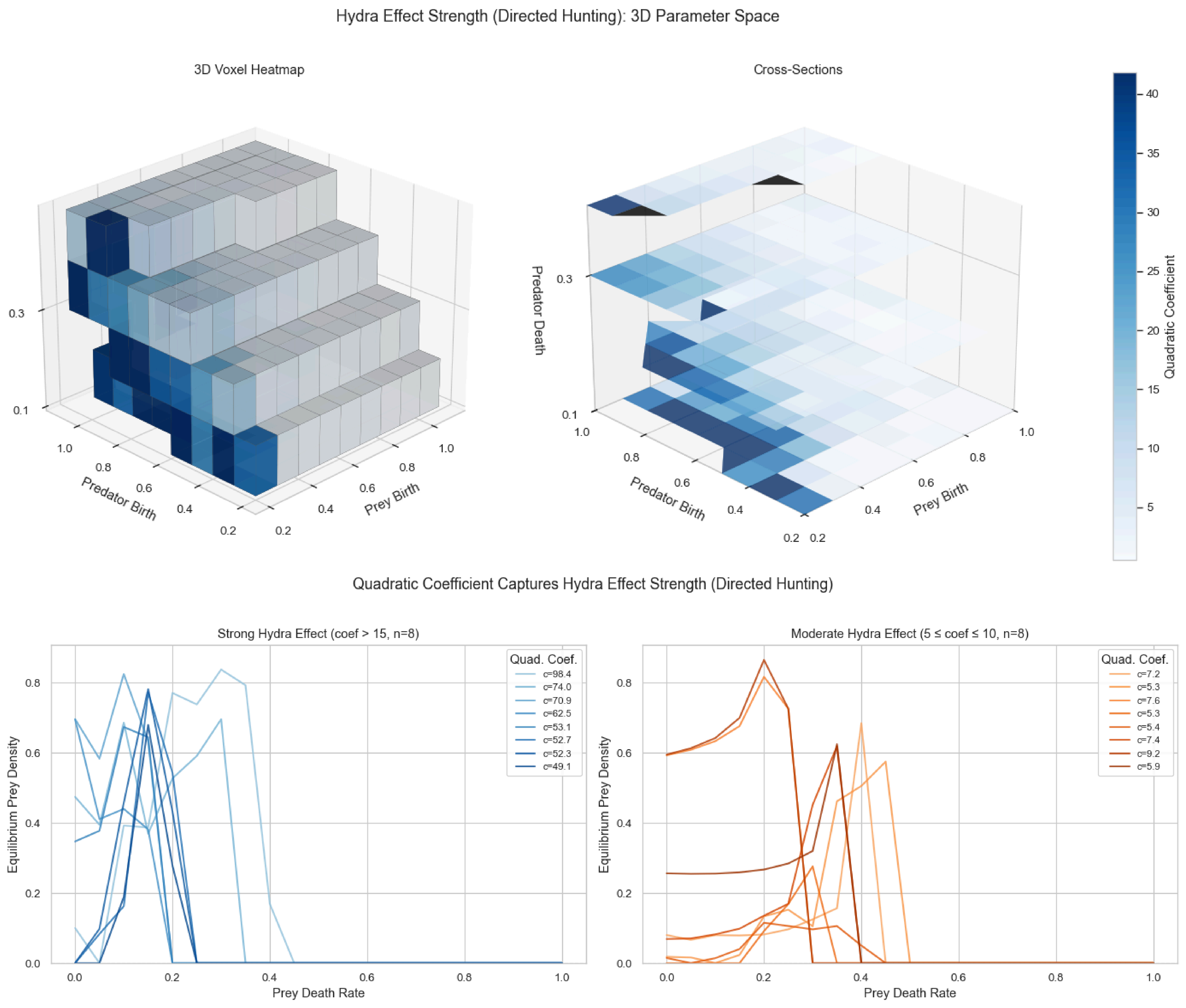
- Parameter sweeps go up to 1 - we only find hydra regions in the cube.
- Computing quadratic coefficient reduces dimensions → 3(D)



Directed sweep

Directed Hunting Mode:

- Prey scans for empty cells for reproduction
- Predator scans neighbours for prey



Evolution

Per-cell evolution:

- Offspring inherits noisy death rate from parent
- Does the population self-organize?

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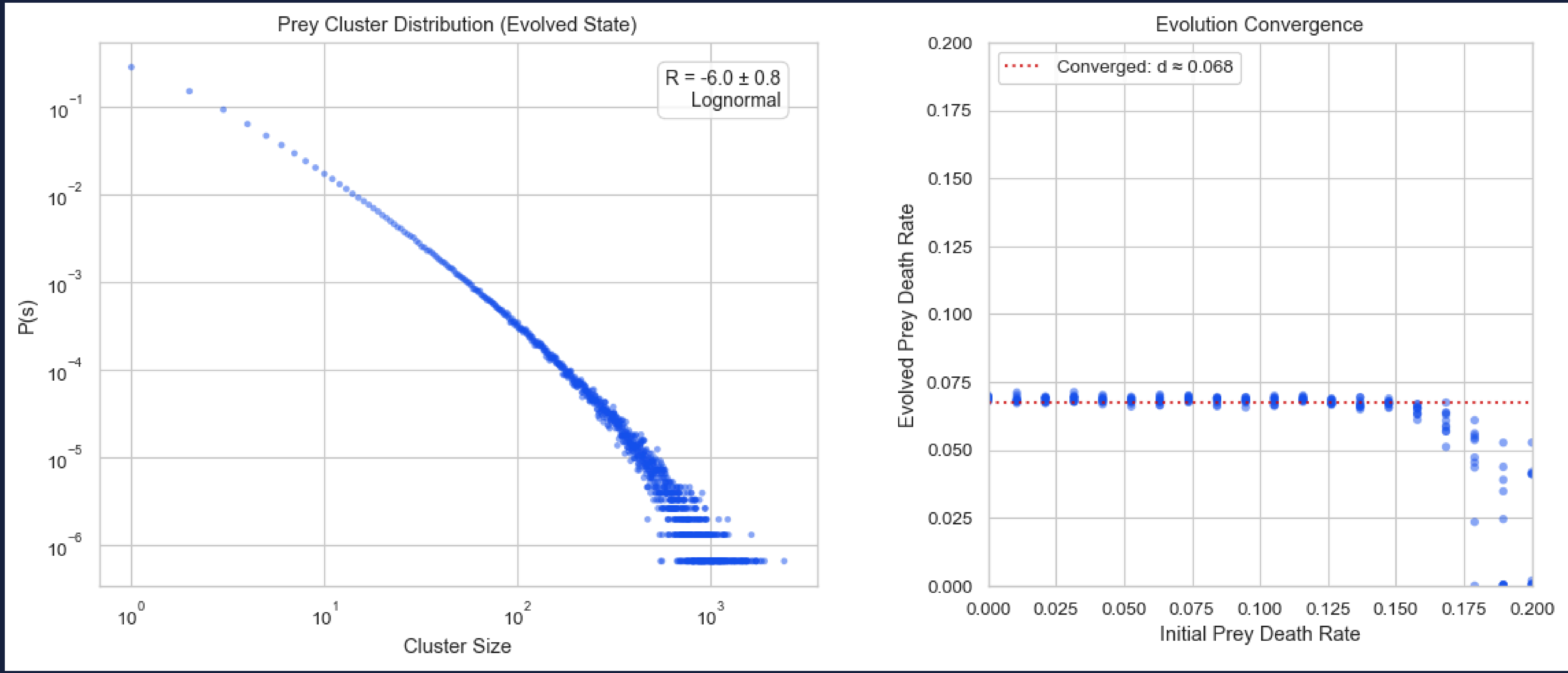

Limitations and Future Work

Limitations:

1. Model lacks active mobility, as agents are restricted to fixed grid positions.
2. Interactions are limited to the immediate n=8 neighbourhood, potentially underestimating predator mobility.

Future Work:

1. Transition from static grid-based reproduction to an agent-based model with active mobility.
2. Modify the model to allow for variable interaction radius, allowing predators to actively hunt prey.
3. Research other power-law mechanics, such as critical slowing down.
4. Study why these parameters result in the hydra effect.
5. Investigate why population converges at death rate of 0.068.

Questions?

GIT REPOSITORY:

@https://github.com/codegithubka/CSS_Project